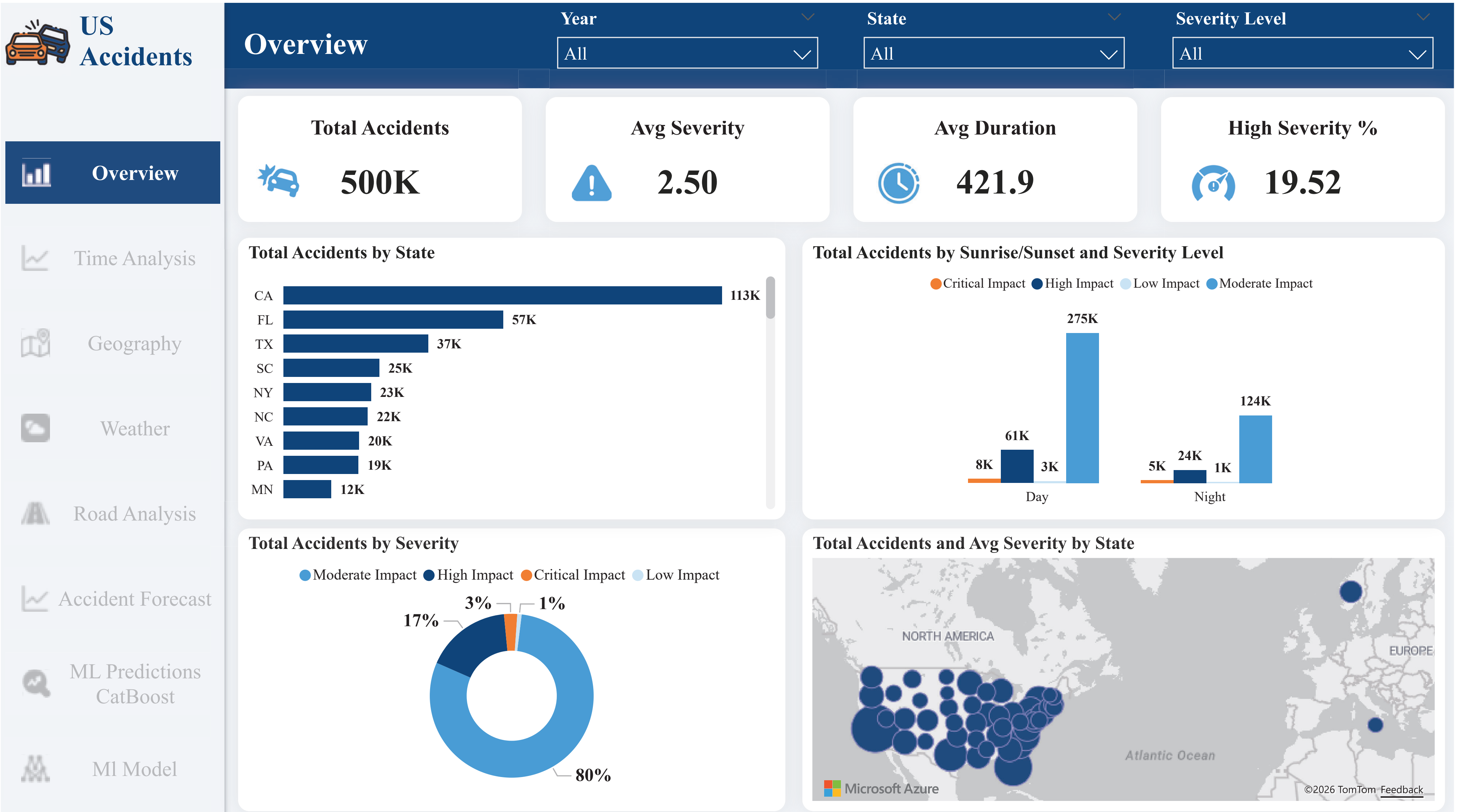

US Accidents Analysis Dashboard
Visualize and track accident data across the USA.
Launch Dashboard
Dashboard Features
Time Analysis
Geography
Weather & Environment
Road Features
Accident Forecast
ML Predictions CatBoost
ML Model



US Accidents
Overview
Time Analysis
Geography
Weather
Road Analysis
Accident Forecast
ML Predictions CatBoost
Ml Model
Overview
Year
All
State
All
Severity Level
All
Total Accidents
500K
Avg Severity
2.50
Avg Duration
421.9
High Severity %
19.52
Total Accidents by State
CA 113K
FL 57K
TX 37K
SC 25K
NY 23K
NC 22K
VA 20K
PA 19K
MN 12K
Total Accidents by Sunrise/Sunset and Severity Level
Critical Impact
High Impact
Low Impact
Moderate Impact
8K
61K
3K
275K
Day
5K
24K
1K
124K
Night
Total Accidents by Severity
Moderate Impact
High Impact
Critical Impact
Low Impact
17%
3%
1%
80%
Total Accidents and Avg Severity by State
NORTH AMERICA
EUROPE
Atlantic Ocean
Microsoft Azure
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Feedback

US Accidents

- Overview
- **Time Analysis**
- Geography
- Weather
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Time Analysis

Year	Month	Severity Level	State
All	All	All	All

Rush Hour Accidents

202K

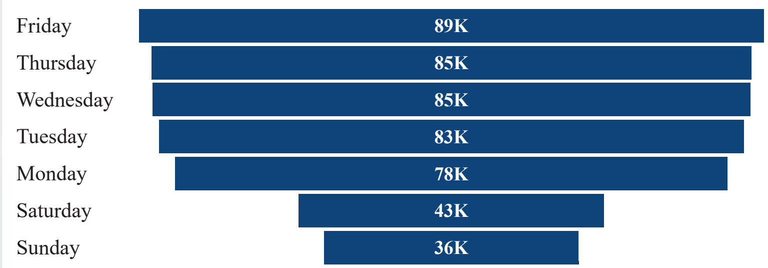
Weekend Accidents

80K

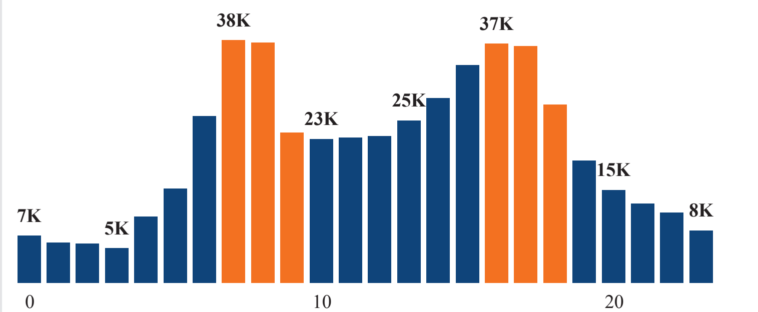
Total Accident by Month and Year

Month	2016	2017	2018	2019	2020	2021	2022	2023
1	2	3586	4874	5189	5909	9868	8813	10312
2	58	3374	4679	4892	5508	10270	10626	3555
3	378	3718	4754	4375	5469	5856	9110	1897
4	1196	3164	4773	4783	5738	6050	12674	
5	1185	2639	4939	4749	5641	6671	10330	
6	1941	2904	4416	4234	6429	8487	8249	
7	3055	2806	4223	4424	2297	7309	9249	
8	3566	5334	4875	4923	2418	7858	9696	
9	3603	5043	4784	5667	5514	8777	8867	
10	3570	4822	5472	7032	8545	8803	5595	
11	4281	4571	5270	5233	11164	10095	8406	
12	3828	4553	4519	6351	11523	11696	12119	

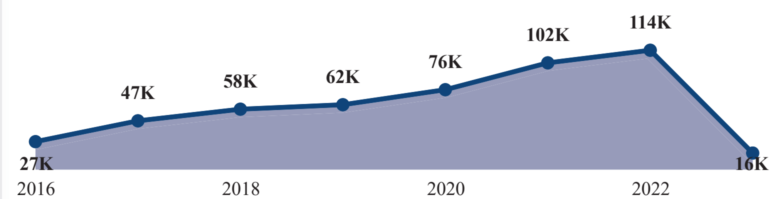
Total Accidents by Day of Week

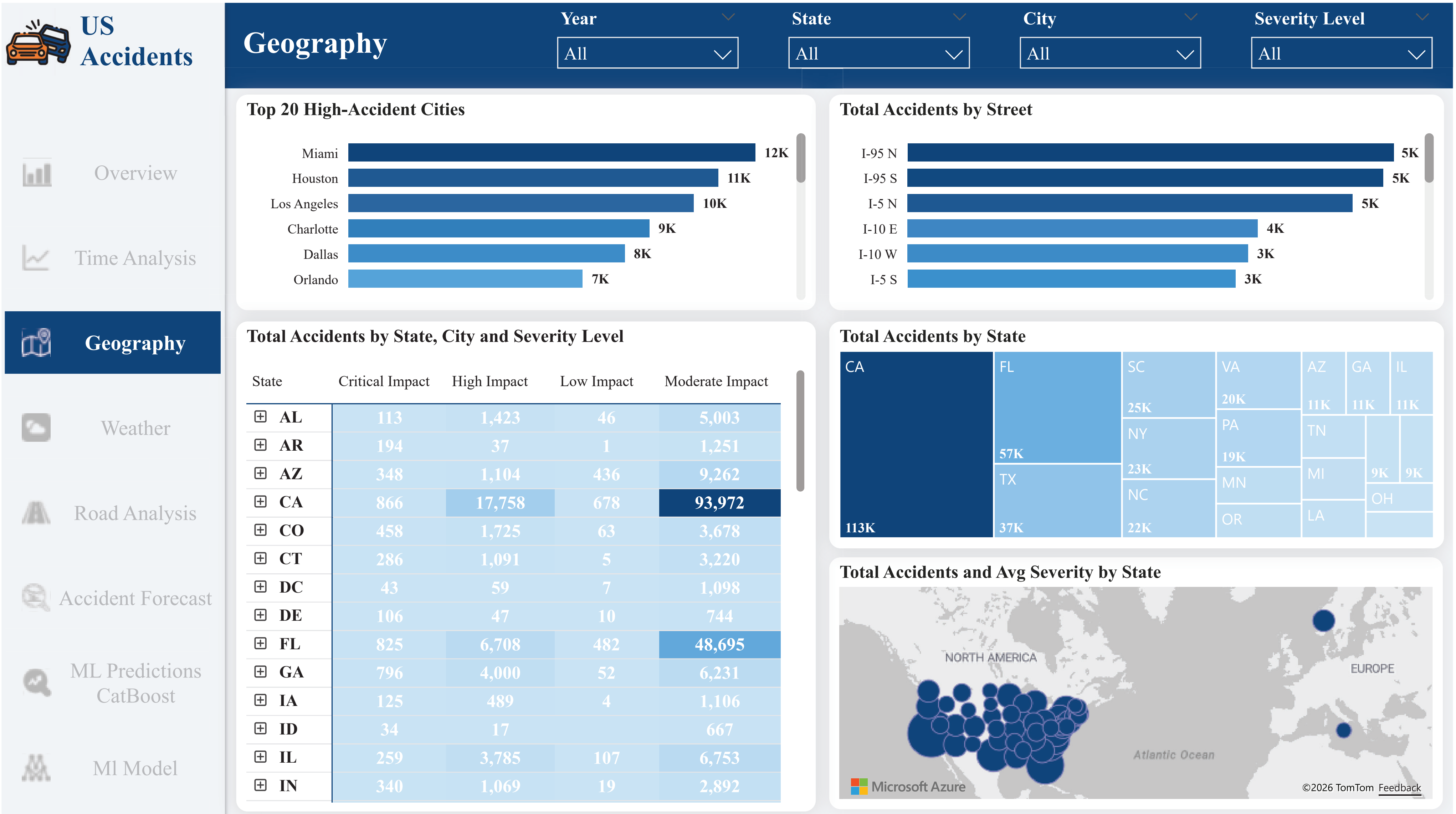


Total Accident by Hour



Total Accidents by Year




US Accidents
Overview
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Geography
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ML Predictions CatBoost
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Geography
Year
All
State
All
City
All
Severity Level
All
Top 20 High-Accident Cities
Miami 12K
Houston 11K
Los Angeles 10K
Charlotte 9K
Dallas 8K
Orlando 7K
Total Accidents by Street
I-95 N 5K
I-95 S 5K
I-5 N 5K
I-10 E 4K
I-10 W 3K
I-5 S 3K
Total Accidents by State, City and Severity Level

State	Critical Impact	High Impact	Low Impact	Moderate Impact
AL	113	1,423	46	5,003
AR	194	37	1	1,251
AZ	348	1,104	436	9,262
CA	866	17,758	678	93,972
CO	458	1,725	63	3,678
CT	286	1,091	5	3,220
DC	43	59	7	1,098
DE	106	47	10	744
FL	825	6,708	482	48,695
GA	796	4,000	52	6,231
IA	125	489	4	1,106
ID	34	17		667
IL	259	3,785	107	6,753
IN	340	1,069	19	2,892

Total Accidents by State
CA 113K
FL 57K
TX 37K
SC 25K
NY 23K
NC 22K
VA 20K
PA 19K
MN
OR
AZ 11K
GA 11K
IL 11K
TN
MI
LA
9K
9K
OH
Total Accidents and Avg Severity by State
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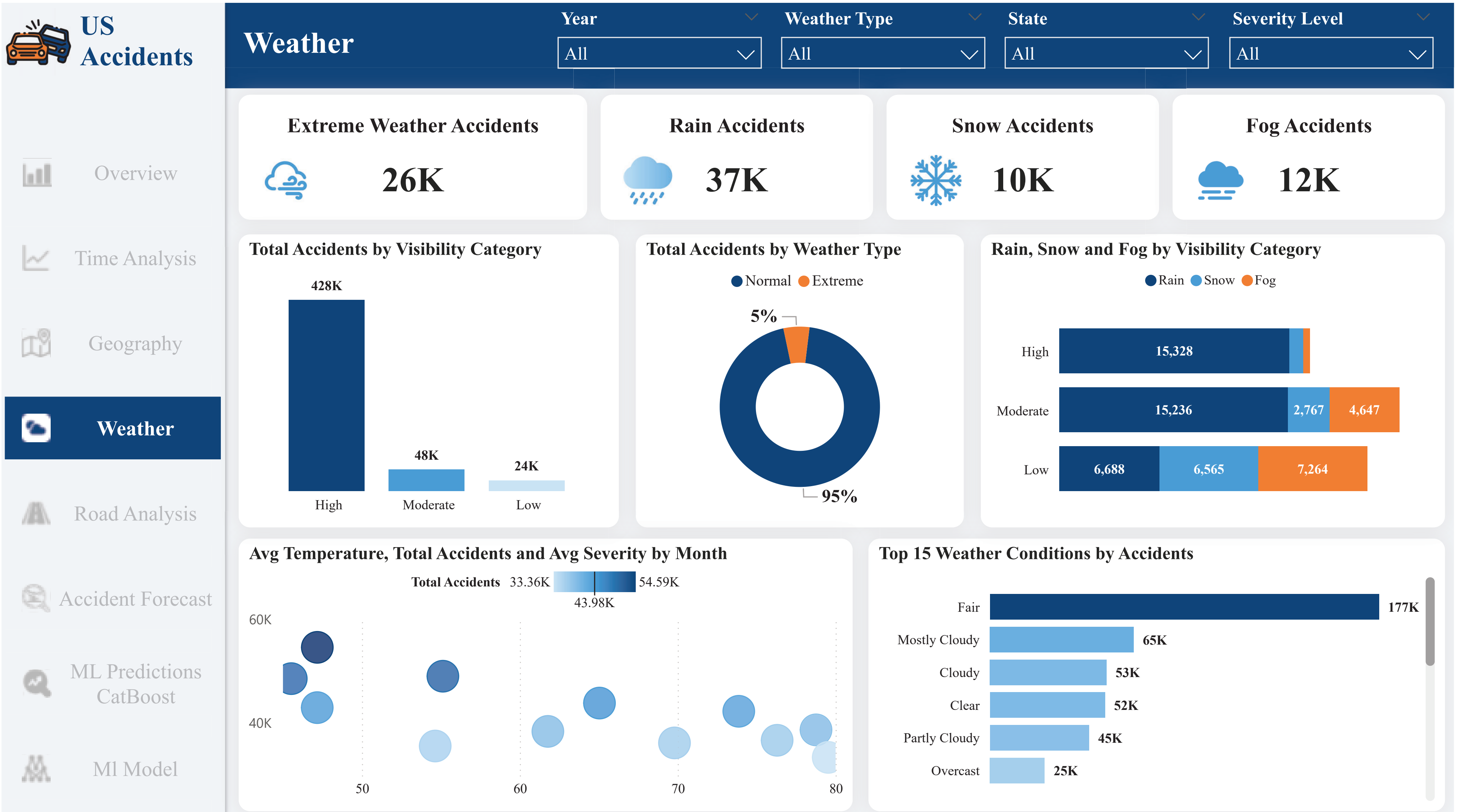
US Accidents
Weather
Year
All
Weather Type
All
State
All
Severity Level
All
Overview
Time Analysis
Geography
Weather
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Extreme Weather Accidents
26K
Rain Accidents
37K
Snow Accidents
10K
Fog Accidents
12K
Total Accidents by Visibility Category
428K
48K
24K
High
Moderate
Low
Total Accidents by Weather Type
Normal
Extreme
5%
95%
Rain, Snow and Fog by Visibility Category
Rain
Snow
Fog
High
15,328
Moderate
15,236
2,767
4,647
Low
6,688
6,565
7,264
Avg Temperature, Total Accidents and Avg Severity by Month
Total Accidents 33.36K 54.59K
43.98K
60K
40K
50
60
70
80
Top 15 Weather Conditions by Accidents
Fair 177K
Mostly Cloudy 65K
Cloudy 53K
Clear 52K
Partly Cloudy 45K
Overcast 25K

US Accidents

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Road Analysis

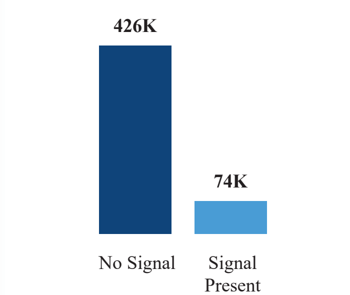
Year	Rush Hour Status	State	Severity Level
All	All	All	All

Signal Accidents	Junction Accidents	Crossing Accidents	Avg Road Features
74K	37K	56K	0.42

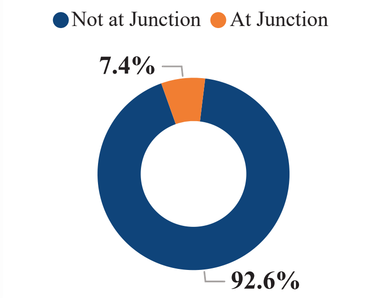
Total Accidents by Road Feature and Severity Level

Road_Feature	Critical Impact	High Impact	Low Impact	Moderate Impact
Amenity	132	276	87	5696
Bump	6	20	2	183
Crossing	816	3185	1171	51081
Give_Way	82	314	40	1926
Junction	1324	8693	208	26782
No_Exit	37	112	25	1071
Railway	93	566	69	3592
Roundabout				13
Station	170	819	174	11835
Stop	358	521	167	13017
Traffic_Calming	10	49	3	404
Traffic_Signal	1263	5812	1669	65291

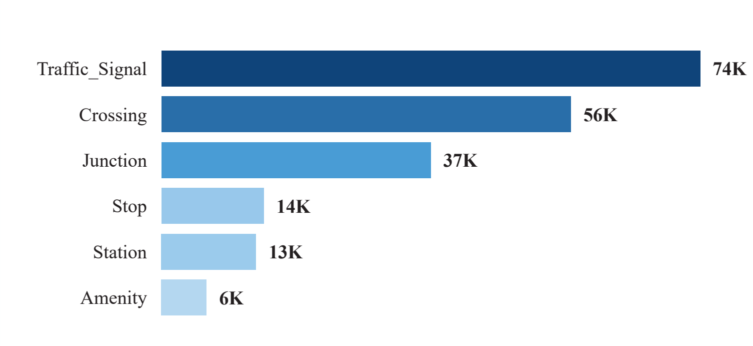
Accidents by Signal Status



Accidents by Junction Status



Accidents by Road Feature Presence



US Accidents

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Accident Forecast

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Accident Forecast

Forecast Data | Historical Data

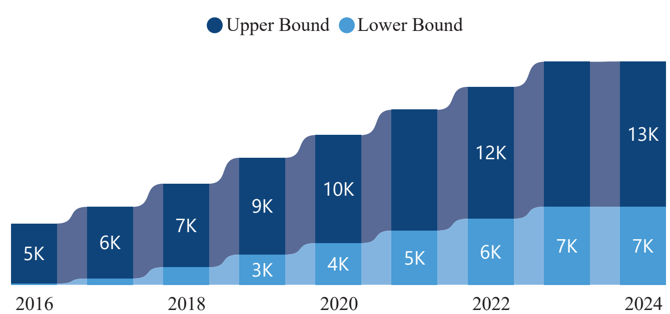
Total Historical Accidents
497K

Total Predicted Next 12 Months
124K

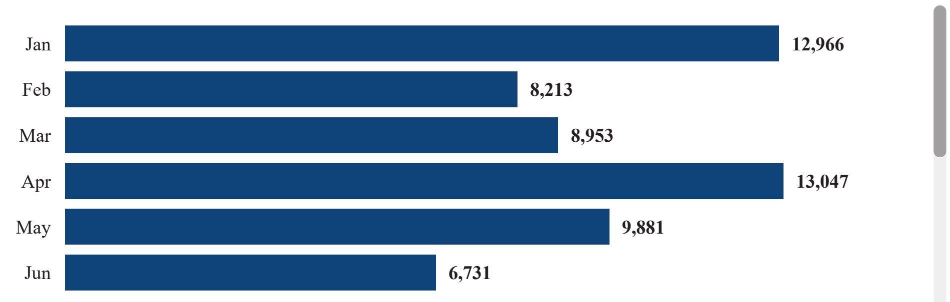
Peak Month Prediction
14K

Lowest Month Prediction
7K

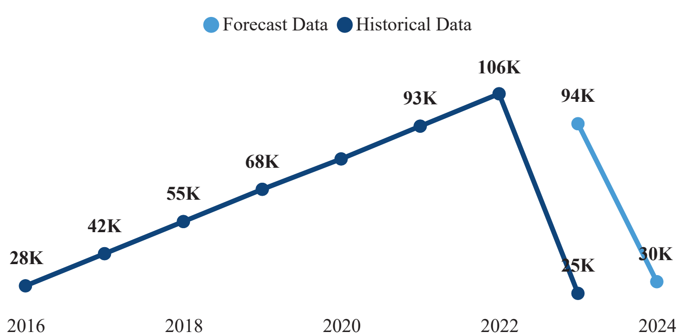
Forecast Confidence Interval (95%)



Predicted Accidents per Month



US Accidents — 12 Month Forecast (Prophet Model)



"Forecast Insights"

- December 2023 predicted as highest risk month: **14,415 accidents**
- June 2023 predicted as lowest risk month: **6,731 accidents**
- Winter months (Dec-Jan) show 40% more accidents than summer
- Model trained on 87 months of historical data (2016-2023)
- 95% confidence interval ensures reliable prediction range"

US Accidents

ML Predictions CatBoost

Risk Level: All | Year: All | Is_Rush_Hour: All

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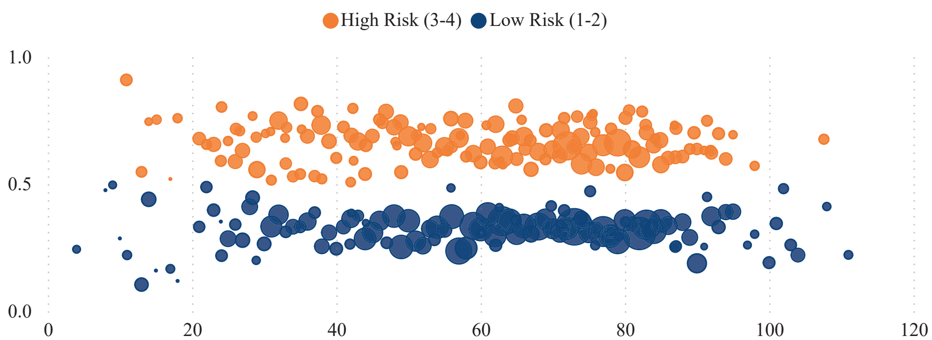
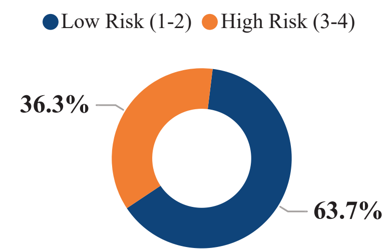
Total Predicted High Risk	Total Predicted Low Risk	Avg Risk Score	High Risk %
363	637	0.44	36.30

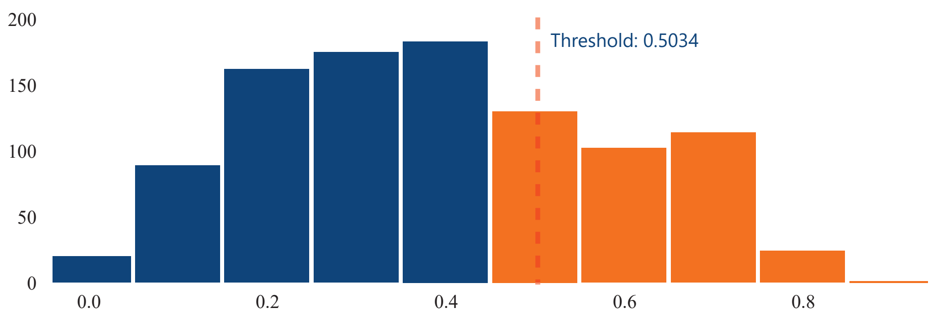
Risk Score and Wind Speed by Predicted Risk Level and Temperature



Predicted Risk Level



Risk Score Prob by Risk Score (Bins)



Year	Weather Condition Encoded	Risk Level	Risk Score
2016	42	High Risk (3-4)	0.79
2016	62	High Risk (3-4)	0.77
2018	42	High Risk (3-4)	0.76
2017	77	High Risk (3-4)	0.76

Model: CatBoost V4 | **Threshold:** 0.5034
Features: 20 | **ROC-AUC:** 0.746

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ML Model

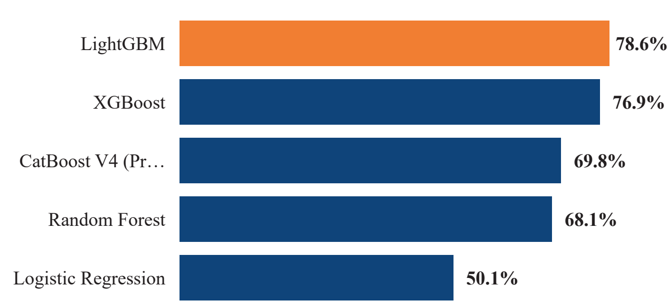
Max of Accuracy

78.6%

Features Count

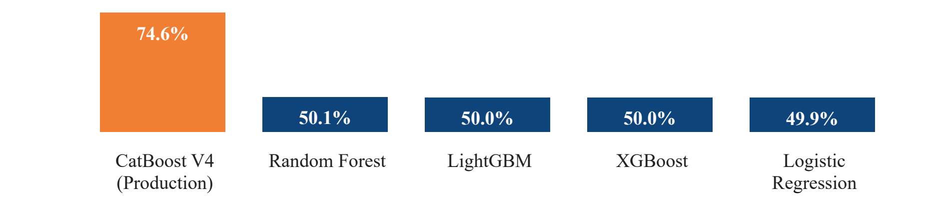
30

Accuracy by Model



Model	Task	Accuracy	ROC_AUC
CatBoost V4 (Production)	Binary	69.76%	74.60%
Random Forest	Binary	68.10%	50.10%
LightGBM	Binary	78.62%	50.00%
XGBoost	Binary	76.85%	50.00%
Logistic Regression	Binary	50.06%	49.90%
LightGBM	Multi	69.83%	
Logistic Regression	Multi	43.94%	
Random Forest	Multi	66.41%	
XGBoost	Multi	68.34%	

ROC_AUC by Model



Importance by Feature

Project Executive Summary

- **Architecture:** Medallion Pipeline (Bronze > Silver > Gold) using Python for ETL and Spark/SQL for data refinement.
- **Data Volume:** Processed and analyzed over **500,000 accident records**, ensuring data quality and integrity throughout the lifecycle.
- **Modeling:** Developed a Multi-class Classification model using **LightGBM**, achieving an optimized **Accuracy of 69.83%**.
- **Best Model**: CatBoost (Binary Classification) — ROC-AUC: 0.746, Threshold: 0.5034.
- **Key Features:** Distance(mi), Wind_Speed_mph, Temperature_F, Pressure_in, Humidity_pct drive predictions.
- **Forecast:** Prophet model trained on 87 months (2016–2023) predicts 124K accidents in next 12 months.